MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY, TTL, OCTAL BUFFER GATES WITH THREE-STATE OUTPUTS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, low-power Schottky, octal bus buffer gates with three-state outputs. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	Inverting octal buffer gate (inverting control inputs)
02	Noninverting octal buffer gate (complementary control inputs)
03	Noninverting octal buffer gates (inverting control inputs)
04	Inverting octal buffer gate (inverting control inputs)
05	Noninverting octal buffer gates (inverting control inputs)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
R	D-8 (20-lead, 1/4" x 1 1/16"), dual-in-line package
S	F-9 (20-lead, 1/4" x 1/2"), flat package
2	C-2 (20 terminal, .350" x .350"), square chip carrier package
X	C-2A (20 terminal, .350" x .350"), square chip carrier package

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V to +7.0 V
Input voltage range	-1.5 V at -18 mA to +5.5 V
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/:	
Device type 01	275 mW
Device types 02 and 03	297 mW
Device type 04	286 mW
Device type 05	302 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases R, S, and X	See MIL-M-38510, appendix C
Case 2	60°C/W 2/
Junction temperature (T_J) 3/	+175°C

1.4 Recommended operating conditions.

Low-level output current	18 mA maximum
High-level output current	-12 mA maximum
Supply voltage (V_{CC})	4.5 V minimum to 5.5 V maximum
Minimum high-level input voltage (V_{IH})	2.0 V
Maximum low-level input voltage (V_{IL})	0.7 V
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specification and standard. Unless otherwise specified, the following specification and standard, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.

3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883.

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3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 9 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D, E, or F (condition F applies to class B devices only) using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$		Device type	Limits Min	Limits Max	Unit
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA, T_C = +25°C		ALL		-1.5	V
High level output voltage	V_{OH1}	V_{CC} = 4.5 V, V_{IH} = 2.0 V, V_{IL} = 0.7 V, I_{OH} = -3 mA		ALL	2.4		V
High level output voltage	V_{OH2}	V_{CC} = 4.5 V, V_{IH} = 2.0 V, V_{IL} = 0.5 V, I_{OH} = -12 mA		ALL	2.0		V
Low level output voltage	V_{OL1}	V_{CC} = 4.5 V, I_{OL} = 12 mA, V_{IH} = 2.0 V, V_{IL} = 0.7 V		ALL		0.4	V
Low level output voltage	V_{OL2}	V_{CC} = 4.5 V, I_{OL} = 18 mA, V_{IH} = 2.0 V, V_{IL} = 0.7 V		ALL		0.45	V
Off-state (high-impedance state) output current	I_{OZH}	V_{CC} = 5.5 V, V_{OH} = 2.7 V, V_{IL} = 0.7 V, V_{IH} = 2.0 V		ALL		20	µA
Off-state (high-impedance state) output current	I_{OZL}	V_{CC} = 5.5 V, V_{OL} = 0.4 V, V_{IL} = 0.7 V, V_{IH} = 2.0 V		ALL		-20	µA
High level input current (all inputs)	I_{IH1}	V_{CC} = 5.5 V, V_{IH} = 2.7 V		ALL		20	µA
High level input current (all inputs)	I_{IH2}	V_{CC} = 5.5 V, V_{IH} = 5.5 V		ALL		100	µA
Low level input current (all inputs)	I_{IL}	V_{CC} = 5.5 V, V_{IL} = 0.4 V		ALL	0	-200	µA
Supply current	I_{CCH}	V_{CC} = 5.5 V	Outputs high	01, 02, 03, 04		27	mA
				05		32	mA
	I_{CCL}		Outputs low	01		44	mA
				02, 03		46	mA
				04		45	mA
				05		52	mA
	I_{CCZ}		Outputs disabled	01		50	mA
				02, 03		54	mA
				04		52	mA
				05		55	
Short circuit output current 1/	I_{OS}	V_{CC} = 5.5 V	ALL	-40	-225	mA	
Propagation delay time (low-to-high level)	t_{PLH1}	C_L = 50 pF, R_L = 110 ohms, V_{CC} = 5.0 V		01,04,05	2	18	ns
				02, 03,	2	23	ns
Propagation delay time (high-to-low level)	t_{PHL1}	C_L = 50 pF, R_L = 110 ohms, V_{CC} = 5.0 V		ALL	2	23	ns
Output enable time to high level	t_{PZH}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 110 ohms		ALL	2	39	ns

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
Output enable time to low level	t_{PZL}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 110 ohms	ALL	2	39	ns
Output disable time from high level	t_{PHZ}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 110 ohms	ALL	2	45	ns
Output disable time from low level	t_{PLZ}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 110 ohms	ALL	2	39	ns

1/ Not more than one output should be shorted at a time.

TABLE II. Electrical test requirements.

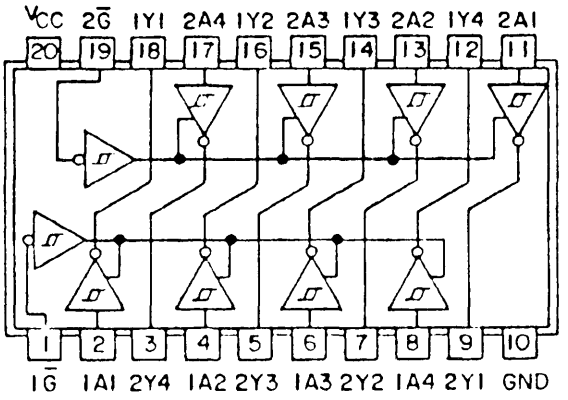
MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7, 9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,	1,2,3,7,9
Group B test requirements (method 5005) subgroup 5	1,2,3,7,9, 10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

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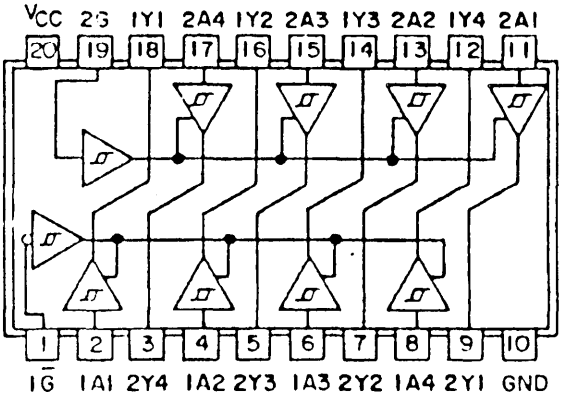
Device type 01

Cases R and S



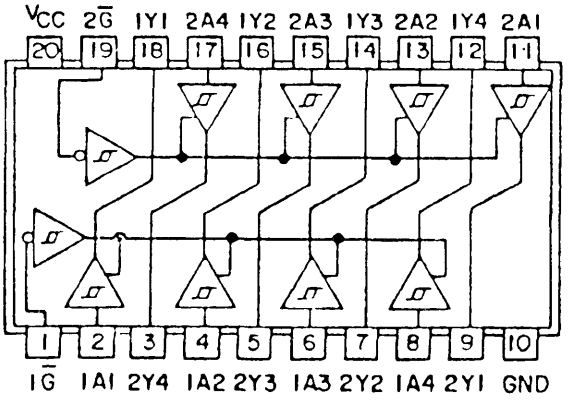
Device type 02

Cases R and S



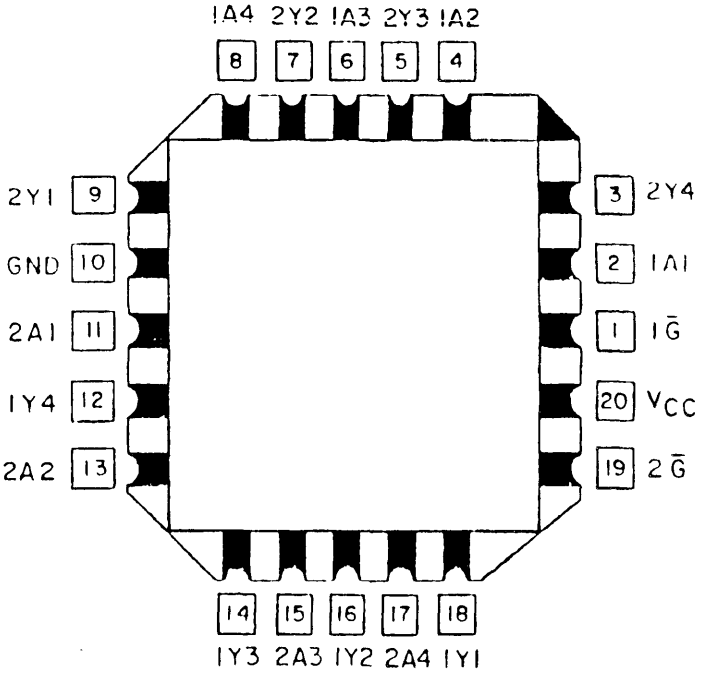
Device type 03

Cases R and S



Device types 01, 02, and 03

Cases 2 and X



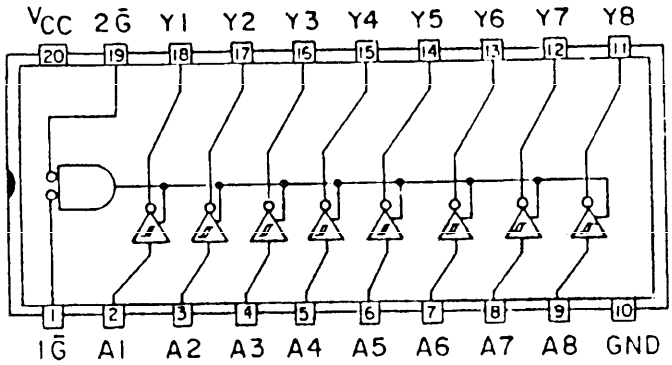
NOTE: For device 02 only, pin 19: 2 G.

FIGURE 1. Logic diagrams and terminal connections (top view).

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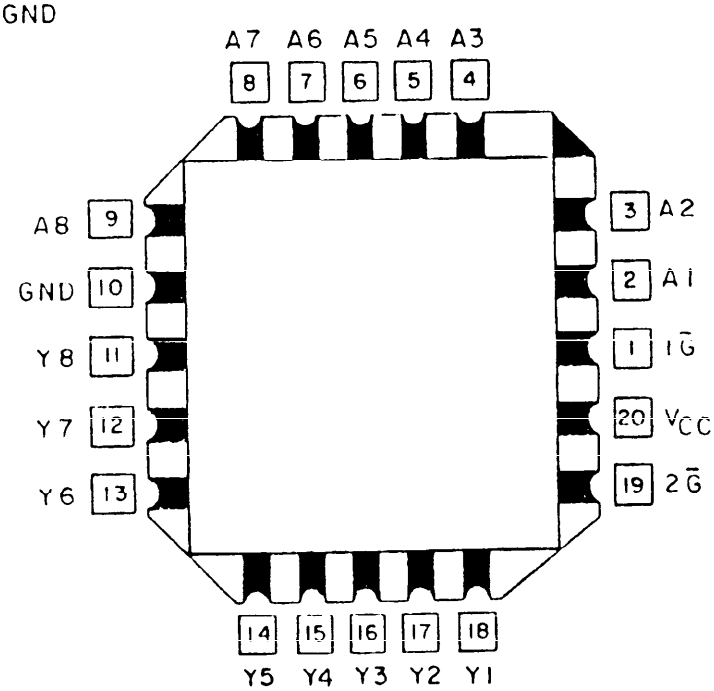
Device type 04

Cases R and S



Device types 04 and 05

Cases 2 and X



(TOP VIEW)

Device type 05

Cases R and S

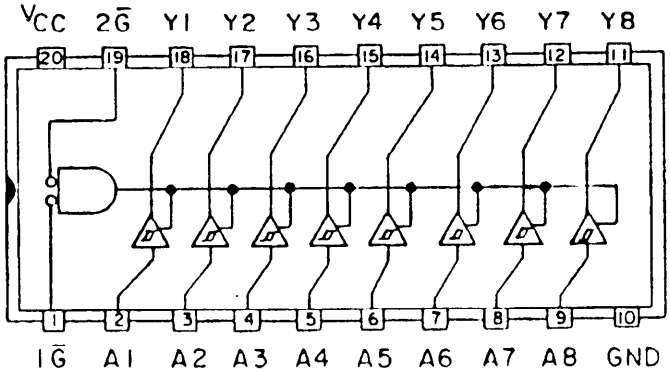


FIGURE 1. Logic diagrams and terminal connections (top view) - Continued.

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Device type 01

Inputs		Out
$\overline{G}$	A	Y
L	H	L
L	L	H
H	X	Z

Device type 02

Inputs				Outputs	
$\overline{1G}$	$\overline{2G}$	1A	2A	$\overline{1Y}$	$\overline{2Y}$
L	H	H	H	H	H
L	H	L	L	L	L
H	L	X	X	Z	Z

Device type 03

Inputs		Out
$\overline{G}$	A	Y
L	L	L
L	H	H
H	X	Z

Device type 04

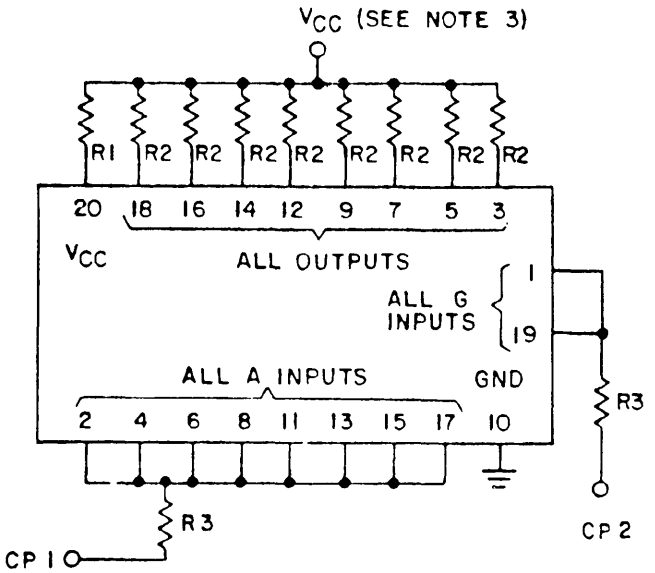
Inputs		OUTPUTS	
$\overline{1G}$	$\overline{2G}$	A	Y
L	L	H	L
L	L	L	H
X	H	X	Z
H	X	X	Z
H	H	X	Z

Device type 05

INPUTS		OUTPUTS	
$\overline{1G}$	$\overline{2G}$	A	Y
L	L	L	L
L	L	H	H
X	H	X	Z
H	X	X	Z
H	H	X	Z

FIGURE 2. Truth tables.

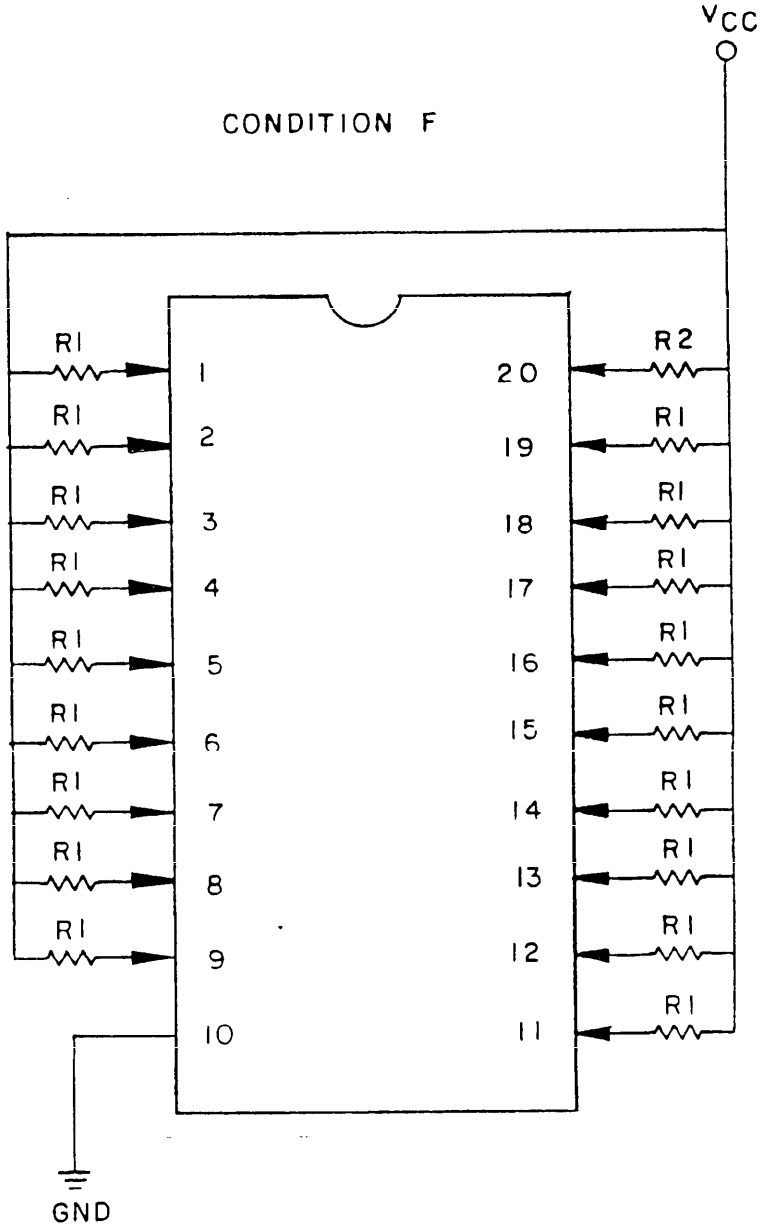
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NOTES:
1. CP1 = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. CP2 = 50 kHz ±50% square wave, otherwise as described in note 1 above.
3. R_1 and V_{CC} shall be chosen so as to insure 5.0 V at the device terminal.
4. R_2 = 390Ω±5%, R_3 = 27Ω ±5%.

FIGURE 3. Burn-in and life-test circuit for device types 01, 02, 03, 04 and 05.

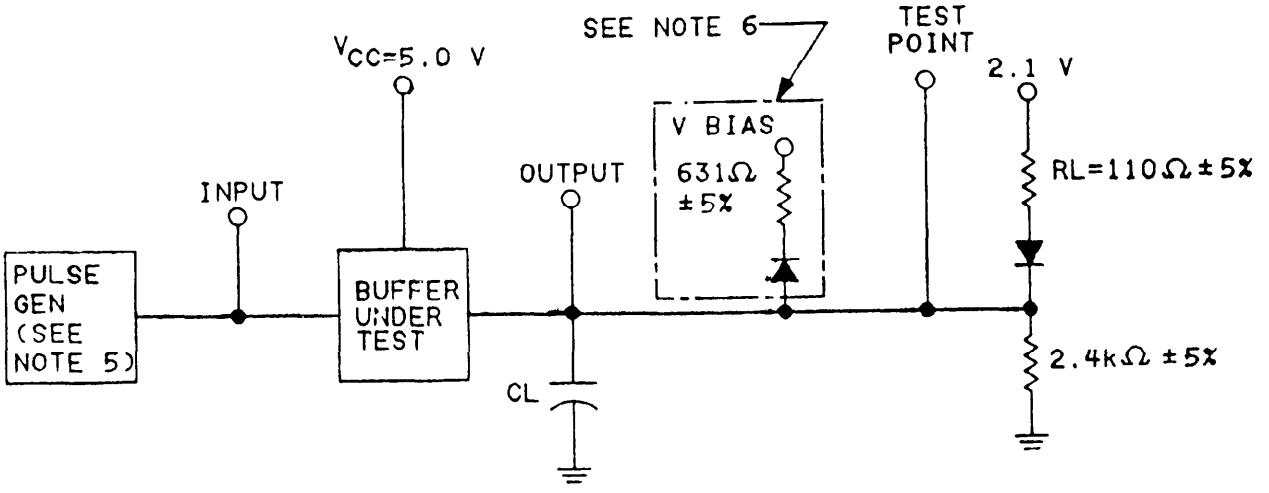
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NOTES:
1. R_1 = 2000 Ω ±10 percent RLC
2. V_{CC} and R2 shall be chosen to insure 5.0 V minimum at the device terminal.

FIGURE 3. Burn-in and life-test circuit for device types 01, 02, 03, 04 and 05 - Continued.

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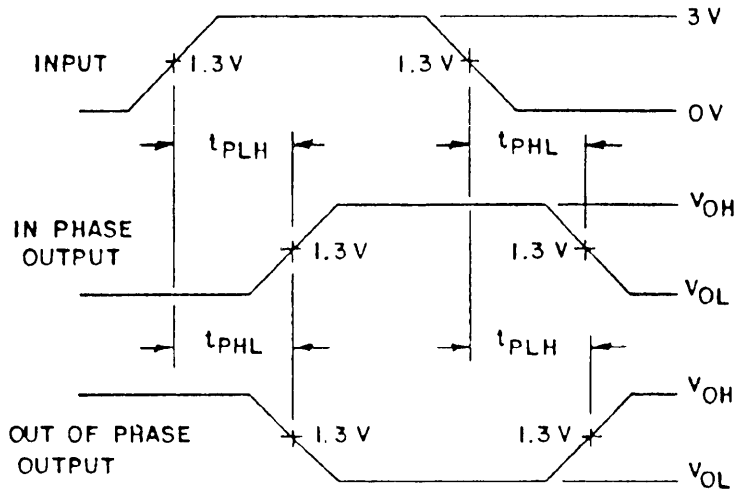


TEST CIRCUIT

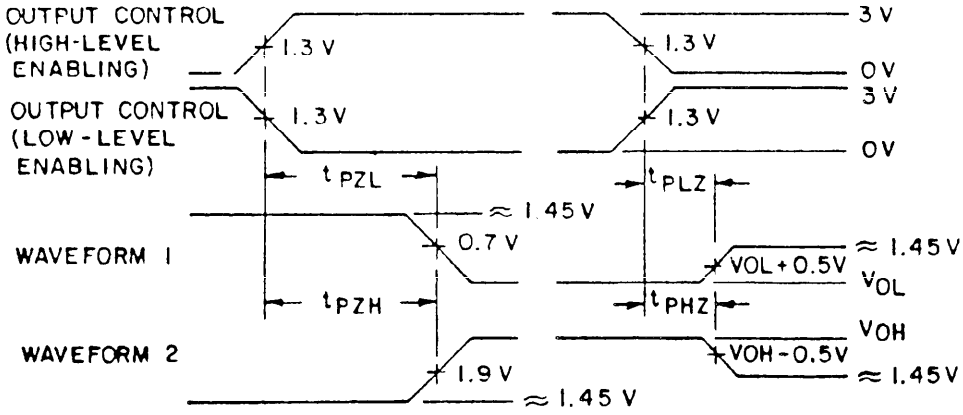
NOTES:
1. C_L = 50 pF ±10%. C_L includes scope probe and jig capacitance.
2. All diodes are 1N3064 or equivalent.
3. Waveform 1 is for an output with internal conditions such that the output is low except when disabled by the output control. Waveform 2 is for an output with internal conditions such that the output is high except when disabled by the output control.
4. In the examples above, the phase relationships between inputs and outputs have been chosen arbitrarily.
5. All input pulses are supplied by generators having the following characteristics: PRR ≤ 1 MHz; t_p = 500 ns; $Z_{OUT} \approx 50$ ohms, V_{GEN} = 3.0 V; $t_{TLH} \leq 15$ ns; and $t_{THL} \leq 6$ ns between 0.7 V and 2.7 V.
6. The diode and resistor shown within the dotted area are optional. When the diode and resistor are used, V_{BIAS} shall be 5.5 V for all tests except for t_{PHZ}; for t_{PHZ} tests, V_{BIAS} shall be -0.6 V.

FIGURE 4. Switching time test circuit and waveforms for device types 01, 02, 03, 04, and 05.

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VOLTAGE WAVEFORMS
PROPAGATION DELAY TIMES



VOLTAGE WAVEFORMS
ENABLE AND DISABLE TIMES, THREE-STATE OUTPUTS

FIGURE 4. Switching time test circuit and waveforms for device types 01, 02, 03, 04, and 05 - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2,X Test no	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				1G	1A1	2Y4	1A2	2Y3	1A3	2Y2	1A4	2Y1	GND	2A1	1Y4	2A2	1Y3	2A3	1Y2	2A4	1Y1	2G	V_{CC}				
1	V_{OH1}	3006	1	0.7 V	0.7 V								GND								-3 mA		4.5 V	1Y1	2.4		V
T_C = +25°C		"	2	"			0.7 V						"						-3 mA				"	1Y2	"		"
		"	3	"					0.7 V				"				-3 mA						"	1Y3	"		"
		"	4	"							0.7 V		"		-3 mA								"	1Y4	"		"
		"	5									-3 mA	"	0.7 V								0.7 V	"	2Y1	"		"
		"	6							-3 mA			"			0.7 V						"	"	2Y2	"		"
		"	7					-3 mA					"					0.7 V					"	"	2Y3	"		"
		"	8			-3 mA							"							0.7 V			"	"	2Y4	"		"
	V_{OH}	"	9	0.5 V	0.5 V								"								-12 mA			"	1Y1	2.0		"
		"	10	"			0.5 V						"						-12 mA				"	1Y2	"		"
		"	11	"					0.5 V				"				-12 mA						"	1Y3	"		"
		"	12	"							0.5 V		"		-12 mA								"	1Y4	"		"
		"	13									-12 mA	"	0.5 V								0.5 V	"	2Y1	"		"
		"	14							-12 mA			"			0.5 V						"	"	2Y2	"		"
		"	15					-12 mA					"					0.5 V					"	"	2Y3	"		"
		"	16			-12 mA							"							0.5 V			"	"	2Y4	"		"
	V_{OL1}	3007	17	0.7 V	2.0 V								"								12 mA			"	1Y1		0.4	"
		"	18	"			2.0 V						"						12 mA				"	1Y2		"	"
		"	19	"					2.0 V				"				12 mA						"	1Y3		"	"
		"	20	"							2.0 V		"		12 mA								"	1Y4		"	"
		"	21									12 mA	"	2.0 V								0.7 V	"	2Y1		"	"
		"	22							12 mA			"			2.0 V						"	"	2Y2		"	"
		"	23					12 mA					"					2.0 V					"	"	2Y3		"	"
		"	24			12 mA							"							2.0 V			"	"	2Y4		"	"
	V_{OL2}	"	25	0.7 V	2.0 V								"								18 mA			"	1Y1		0.45	"
		"	26	"			2.0 V						"						18 mA				"	1Y2		"	"
		"	27	"					2.0 V				"				18 mA						"	1Y3		"	"
		"	28	"							2.0 V		"		18 mA								"	1Y4		"	"
		"	29									18 mA	"	2.0 V								0.7 V	"	2Y1		"	"
		"	30							18 mA			"			2.0 V						"	"	2Y2		"	"
		"	31					18 mA					"					2.0 V					"	"	2Y3		"	"
		"	32			18 mA							"							2.0 V			"	"	2Y4		"	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2,X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				1G	1A1	2Y4	1A2	2Y3	1A3	2Y2	1A4	2Y1	GND	2A1	1Y4	2A2	1Y3	2A3	1Y2	2A4	1Y1	2G	V_{CC}				
1	I_{OZH}		33	2.0 V									GND								2.7 V		5.5 V	1Y1		20	µA
T_C = +25°C			34	"									"						2.7 V				"	1Y2		"	"
			35	"									"				2.7 V						"	1Y3		"	"
			36	"									"		2.7 V								"	1Y4		"	"
			37									2.7 V	"									2.0 V	"	2Y1		"	"
			38							2.7 V			"									"	"	2Y2		"	"
			39					2.7 V					"									"	"	2Y3		"	"
			40			2.7 V							"									"	"	2Y4		"	"
	I_{OZL}		41	2.0 V									"								0.4 V	5.5 V	1Y1		-20	"
			42	"									"						0.4 V			"	1Y2		"	"
			43	"									"				0.4 V					"	1Y3		"	"
			44	"									"		0.4 V							"	1Y4		"	"
			45									0.4 V	"									2.0 V	2Y1		"	"
			46							0.4 V			"									"	2Y2		"	"
			47					0.4 V					"									"	2Y3		"	"
			48			0.4 V							"									"	2Y4		"	"
	I_{IL}	3009	49	0.4 V									"										1G	1/	1/	"
		"	50		0.4 V								"										1A1	"	"	"
		"	51				0.4 V						"										1A2	"	"	"
		"	52						0.4 V				"										1A3	"	"	"
		"	53								0.4 V		"										1A4	"	"	"
		"	54										"	0.4 V									2A1	"	"	"
		"	55										"			0.4 V							2A2	"	"	"
		"	56										"					0.4 V					2A3	"	"	"
		"	57										"							0.4 V			2A4	"	"	"
		"	58										"									0.4 V	2G	"	"	"
	I_{IH1}	3010	59	2.7 V									"										1G		20	"
		"	60		2.7 V								"										1A1		"	"
		"	61				2.7 V						"										1A2		"	"
		"	62						2.7 V				"										1A3		"	"
		"	63								2.7 V		"										1A4		"	"
		"	64										"	2.7 V									2A1		"	"

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TABLE III. Group A inspection for device type 01 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases C, D, 2, X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				1A1	2Y4		1A2	2Y3	1A3	2Y2	1A4	2Y1	GND	2A1	1Y4	2A2	1Y3	2A3	1Y2	2A4	1Y1	2$\overline{G}$	V_{CC}				
1	I_{IH1}	3010	65										GND			2.7 V							5.5 V	2A2		20	µA
T_C = +25°C		"	66										"					2.7 V					"	2A3		"	"
		"	67										"							2.7 V			"	2A4		"	"
		"	68										"									2.7 V	"	2$\overline{G}$		"	"
	I_{IH2}	"	69	5.5 V									"										"	1$\overline{G}$		100	"
		"	70		5.5 V								"										"	1A1		"	"
		"	71				5.5 V						"										"	1A2		"	"
		"	72						5.5 V				"										"	1A3		"	"
		"	73								5.5 V		"										"	1A4		"	"
		"	74										"	5.5 V									"	2A1		"	"
		"	75										"			5.5 V							"	2A2		"	"
		"	76										"					5.5 V					"	2A3		"	"
		"	77										"							5.5 V			"	2A4		"	"
		"	78										"									5.5 V	"	2$\overline{G}$		"	"
	I_{OS}	3011	79	GND	GND								"				GND				GND		"	1Y1	-40	-225	mA
		"	80	"			GND						"						GND				"	1Y2	"	"	"
		"	81	"					GND				"				GND						"	1Y3	"	"	"
		"	82	"							GND		"		GND								"	1Y4	"	"	"
		"	83									GND	"	GND								GND	"	2Y1	"	"	"
		"	84							GND			"			GND						"	"	2Y2	"	"	"
		"	85										"					GND				"	"	2Y3	"	"	"
		"	86			GND							"							GND		"	"	2Y4	"	"	"
	I_{CC}	3005	87	GND	GND		GND		GND		GND		"	GND		GND		GND		GND		"	"	V_{CC}		27	"
	I_{CC1}	"	88	GND	5.5 V		5.5 V		5.5 V		5.5 V		"	5.5 V		5.5 V		5.5 V		5.5 V		"	"	V_{CC}		44	"
	I_{CC2}	"	89	5.5 V									"									5.5 V	"	V_{CC}		50	"
	V_{IC}		90	-18 mA									"										4.5 V	1$\overline{G}$		-1.5	V
			91		18 mA								"										"	1A1		"	"
			92				-18 mA						"										"	1A2		"	"
			93						-18 mA				"										"	1A3		"	"
			94								-18 mA		"										"	1A4		"	"
			95										"	-18 mA									"	2A1		"	"
			96										"			-18 mA							"	2A2		"	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD 883 method	Cases R,S,2,X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$1\overline{G}$	1A1	2Y4	1A2	2Y3	1A3	2Y2	1A4	2Y1	GND	2A1	1Y4	2A2	1Y3	2A3	1Y2	2A4	1Y1	$2\overline{G}$	V_{CC}				
1	V_{IC}		97										GND					-18 mA					4.5 V	2A3		-1.5	V
T_C = +25°C			98										"							-18 mA			"	2A4		-1.5	"
			99										"									-18 mA	"	$2\overline{G}$		-1.5	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																										
7 2/	Truth table tests	3014	100	B	B	H	B	H	B	H	B	H	GND	B	H	B	H	B	H	B	H	B	5.0 V	3/			
T_C = +25°C		3014	101	B	A	L	A	L	A	L	A	L	GND	A	L	A	L	A	L	B	L	B	5.0 V				
9	t_{PLH1}	3003 See fig. 4	102	GND	IN								GND								OUT		5.0 V	1A1 to 1Y1	2	14	ns
T_C = +25°C			103	"			IN						"						OUT				"	1A2 to 1Y2	"	"	"
			104	"					IN				"				OUT						"	1A3 to 1Y3	"	"	"
			105	"							IN		"		OUT								"	1A4 to 1Y4	"	"	"
			106									OUT	"	IN								GND	"	2A1 to 2Y1	"	"	"
			107							OUT			"			IN						"	"	2A2 to 2Y2	"	"	"
			108					OUT					"					IN				"	"	2A3 to 2Y3	"	"	"
			109			OUT							"							IN		"	"	2A4 to 2Y4	"	"	"
	t_{PHL1}	"	110	GND	IN								"								OUT		"	1A1 to 1Y1	"	18	"
		"	111	"			IN						"						OUT				"	1A2 to 1Y2	"	"	"
		"	112	"					IN				"				OUT						"	1A3 to 1Y3	"	"	"
		"	113	"							IN		"		OUT								"	1A4 to 1Y4	"	"	"
		"	114									OUT	"	IN								GND	"	2A1 to 2Y1	"	"	"
		"	115							OUT			"			IN						"	"	2A2 to 2Y2	"	"	"
		"	116					OUT					"					IN				"	"	2A3 to 2Y3	"	"	"
		"	117			OUT							"							IN		"	"	2A4 to 2Y4	"	"	"
	t_{PZL}	"	118	IN	4.5 V								"								OUT		"	$1\overline{G}$ to 1Y1	"	30	"
		"	119	"			4.5 V						"						OUT				"	$1\overline{G}$ to 1Y2	"	"	"
		"	120	"					4.5 V				"				OUT						"	$1\overline{G}$ to 1Y3	"	"	"
		"	121	"							4.5 V		"		OUT								"	$1\overline{G}$ to 1Y4	"	"	"
		"	122									OUT	"	4.5 V								IN	"	$2\overline{G}$ to 2Y1	"	"	"
		"	123							OUT			"			4.5 V						"	"	$2\overline{G}$ to 2Y2	"	"	"

See footnotes at end of device type 01

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Case R,S,2,X1 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$1\overline{G}$	1A1	2Y4	1A2	2Y3	1A3	2Y2	1A4	2Y1	GND	2A1	1Y4	2A2	1Y3	2A3	1Y2	2A4	1Y1	$2\overline{G}$	V_{CC}				
9	t_{PZL}	3003	124					OUT					GND					4.5 V				IN	5.0 V	$2\overline{G}$ to 2Y3	2	30	ns
T_C = +25°C		See fig. 4	125			OUT							"							4.5 V		IN	"	$2\overline{G}$ to 2Y4	"	"	"
	t_{PZH}	"	126	IN	GND								"								OUT		"	$1\overline{G}$ to 1Y1	"	"	"
		"	127	"			GND						"						OUT				"	$1\overline{G}$ to 1Y2	"	"	"
		"	128	"					GND				"				OUT						"	$1\overline{G}$ to 1Y3	"	"	"
		"	129	"							GND		"		OUT								"	$1\overline{G}$ to 1Y4	"	"	"
		"	130									OUT	"	GND								IN	"	$2\overline{G}$ to 2Y1	"	"	"
		"	131							OUT			"			GND						"	"	$2\overline{G}$ to 2Y2	"	"	"
		"	132					OUT					"					GND				"	"	$2\overline{G}$ to 2Y3	"	"	"
		"	133			OUT							"							GND		"	"	$2\overline{G}$ to 2Y4	"	"	"
	t_{PLZ}	"	134	IN	4.5 V								"								OUT		"	$1\overline{G}$ to 1Y1	"	"	"
		"	135	"			4.5 V						"						OUT				"	$1\overline{G}$ to 1Y2	"	"	"
		"	136	"					4.5 V				"				OUT						"	$1\overline{G}$ to 1Y3	"	"	"
		"	137	"							4.5 V		"		OUT								"	$1\overline{G}$ to 1Y4	"	"	"
		"	138									OUT	"	4.5 V								IN	"	$2\overline{G}$ to 2Y1	"	"	"
		"	139							OUT			"			4.5 V						"	"	$2\overline{G}$ to 2Y2	"	"	"
		"	140					OUT					"					4.5 V				"	"	$2\overline{G}$ to 2Y3	"	"	"
		"	141			OUT							"							4.5 V		"	"	$2\overline{G}$ to 2Y4	"	"	"
	t_{PHZ}	"	142	IN	GND								"								OUT		"	$1\overline{G}$ to 1Y1	"	35	"
		"	143	"			GND						"						OUT				"	$1\overline{G}$ to 1Y2	"	"	"
		"	144	"					GND				"				OUT						"	$1\overline{G}$ to 1Y3	"	"	"
		"	145	"							GND		"		OUT								"	$1\overline{G}$ to 1Y4	"	"	"
		"	146									OUT	"	GND								IN	"	$2\overline{G}$ to 2Y1	"	"	"
		"	147							OUT			"			GND						"	"	$2\overline{G}$ to 2Y2	"	"	"
		"	148					OUT					"					GND				"	"	$2\overline{G}$ to 2Y3	"	"	"
		"	149			OUT							"							GND		"	"	$2\overline{G}$ to 2Y4	"	"	"
10	t_{PLH1}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																							"	18	"
T_C = +125°C	t_{PH1}																								"	23	"
	t_{PZL}																								"	39	"
	t_{PZH}																								"	39	"
	t_{PL7}																								"	39	"
	t_{PH7}																								"	45	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2,X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{1G}$	1A1	2Y4	1A2	2Y3	1A3	2Y2	1A4	2Y1	GND	2A1	1Y4	2A2	1Y3	2A3	1Y2	2A4	1Y1	$\overline{2G}$	V_{CC}				
11																											
T_C = -55°C	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C																										

1/ I_{IL} Limits shall be as follows:

Test	Min/Max limits (μA) for circuit A	B	C	D	E	F	S
I_{IL}	-5/-200	0/-150	0/-150	0/-150	-0/-200	-5/-200	0/-150

2/ A = 3.0 V min; B = 0.0 V or GND.

3/ H >1.5 V; L <1.5 V.

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TABLE III. Group A inspection for device types 02 and 03.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2,X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				1G	1A1	2Y4	1A2	2Y3	1A3	2Y2	1A4	2Y1	GND	2A1	1Y4	2A2	1Y3	2A3	1Y2	2A4	1Y1	2G 1/	V_{CC}				
1	V_{OH1}	3007	1	0.7 V	2.0 V								GND								-3 mA		4.5 V	1Y1	2.4		V
T_C = +25°C		"	2	"			2.0 V						"						-3 mA				"	1Y2	"		"
		"	3	"					2.0 V				"				-3 mA						"	1Y3	"		"
		"	4	"							2.0 V		"		-3 mA								"	1Y4	"		"
		"	5									-3 mA	"	2.0 V								2/	"	2Y1	"		"
		"	6							-3 mA			"			2.0 V							"	2Y2	"		"
		"	7					-3 mA					"					2.0 V					"	"	2Y3	"		"
		"	8			-3 mA							"							2.0 V			"	"	2Y4	"		"
	V_{OH2}	"	9	0.5 V	2.0 V								"								-12 mA			"	1Y1	2.0		"
		"	10	"			2.0 V						"						-12 mA				"	1Y2	"		"
		"	11	"					2.0 V				"				-12 mA						"	1Y3	"		"
		"	12	"							2.0 V		"		-12 mA								"	1Y4	"		"
		"	13									-12 mA	"	2.0 V								3/	"	2Y1	"		"
		"	14							-12 mA			"			2.0 V							"	2Y2	"		"
		"	15					-12 mA					"					2.0 V					"	"	2Y3	"		"
		"	16			-12 mA							"							2.0 V			"	"	2Y4	"		"
	V_{OL1}	3006	17	0.7 V	0.7 V								"								12 mA			"	1Y1		0.4	"
		"	18	"			0.7 V						"						12 mA				"	1Y2		"	"
		"	19	"					0.7 V				"				12 mA						"	1Y3		"	"
		"	20	"							0.7 V		"		12 mA								"	1Y4		"	"
		"	21									12 mA	"	0.7 V								2/	"	2Y1		"	"
		"	22							12 mA			"			0.7 V							"	2Y2		"	"
		"	23					12 mA					"					0.7 V					"	"	2Y3		"	"
		"	24			12 mA							"							0.7 V			"	"	2Y4		"	"
	V_{OL2}	"	25	0.7 V	0.7 V								"								18 mA			"	1Y1		0.45	"
		"	26	"			0.7 V						"						18 mA				"	1Y2		"	"
		"	27	"					0.7 V				"				18 mA						"	1Y3		"	"
		"	28	"							0.7 V		"		18 mA								"	1Y4		"	"
		"	29									18 mA	"	0.7 V								2/	"	2Y1		"	"
		"	30							18 mA			"			0.7 V							"	2Y2		"	"
		"	31					18 mA					"					0.7 V					"	"	2Y3		"	"
		"	32			18 mA							"							0.7 V			"	"	2Y4		"	"

See footnotes at end of device types 02 and 03.

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TABLE III. Group A inspection for device types 02 and 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2,X Test no.	1 1G	2 1A1	3 2Y4	4 1A2	5 2Y3	6 1A3	7 2Y2	8 1A4	9 2Y1	10 GND	11 2A1	12 1Y4	13 2A2	14 1Y3	15 2A3	16 1Y2	17 2A4	18 1Y1	19 2G 1/	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1	I_{OZH}		33	2.0 V									GND								2.7 V		5.5 V	1Y1		20	µA
T_C = +25°C			34	"									"						2.7 V				"	1Y2		"	"
			35	"									"				2.7 V						"	1Y3		"	"
			36	"									"		2.7 V								"	1Y4		"	"
			37									2.7 V	"										"	2Y1		"	"
			38							2.7 V			"									4/	"	2Y2		"	"
			39					2.7 V					"									"	"	2Y3		"	"
			40			2.7 V							"									"	"	2Y4		"	"
	I_{OZL}		41	2.0 V									"							0.4 V			"	1Y1		-20	"
			42	"									"						0.4 V				"	1Y2		"	"
			43	"									"				0.4 V						"	1Y3		"	"
			44	"									"		0.4 V								"	1Y4		"	"
			45									0.4 V	"									4/	"	2Y1		"	"
			46							0.4 V			"									"	"	2Y2		"	"
			47					0.4 V					"									"	"	2Y3		"	"
			48			0.4 V							"									"	"	2Y4		"	"
	I_{IL}	3009	49	0.4 V									"										"	1G	5/	5/	"
		"	50		0.4 V								"										"	1A1	"	"	"
		"	51				0.4 V						"										"	1A2	"	"	"
		"	52						0.4 V				"										"	1A3	"	"	"
		"	53								0.4 V		"										"	1A4	"	"	"
		"	54										"	0.4 V									"	2A1	"	"	"
		"	55										"			0.4 V							"	2A2	"	"	"
		"	56										"					0.4 V					"	2A3	"	"	"
		"	57										"							0.4 V			"	2A4	"	"	"
		"	58										"									0.4 V	"	2G	"	"	"
	I_{IH1}	3010	59	2.7 V									"										"	1G	"	20	"
		3010	60		2.7 V								"										"	1A1	"	20	"

See footnotes at end of device types 02 and 03.

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TABLE III. Group A inspection for device types 02 and 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open)

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2,X Test no.	1 1G	2 1A1	3 2Y4	4 1A2	5 2Y3	6 1A3	7 2Y2	8 1A4	9 2Y1	10 GND	11 2A1	12 1Y4	13 2A2	14 1Y3	15 2A3	16 1Y2	17 2A4	18 1Y1	19 2G 1/	20 V_{CC}	Measured terminal	Test limits Min	Max	Unit
1	I_{IH1}	3010	61				2.7 V						GND										5.5 V	1A2		20	uA
T_C = +25°C		"	62						2.7 V				"										"	1A3		"	"
		"	63								2.7 V		"										"	1A4		"	"
		"	64										"	2.7 V									"	2A1		"	"
		"	65										"			2.7 V							"	2A2		"	"
		"	66										"					2.7 V					"	2A3		"	"
		"	67										"							2.7 V			"	2A4		"	"
		"	68										"									2.7 V	"	2G		"	"
	I_{IH2}	"	69	5.5 V									"										"	1G		100	"
		"	70		5.5 V								"										"	1A1		"	"
		"	71				5.5 V						"										"	1A2		"	"
		"	72						5.5 V				"										"	1A3		"	"
		"	73								5.5 V		"										"	1A4		"	"
		"	74										"	5.5 V									"	2A1		"	"
		"	75										"			5.5 V							"	2A2		"	"
		"	76										"					5.5 V					"	2A3		"	"
		"	77										"							5.5 V			"	2A4		"	"
		"	78										"									5.5 V	"	2G		"	"
	I_{OS}	3011	79	GND	5.5 V								"								GND		"	1Y1	-40	-225	mA
		"	80	"			5.5 V						"						GND				"	1Y2	"	"	"
		"	81	"					5.5 V				"				GND						"	1Y3	"	"	"
		"	82	"							5.5 V		"		GND								"	1Y4	"	"	"
		"	83									GND	"	5.5 V							6/		"	2Y1	"	"	"
		"	84							GND			"			5.5 V					"		"	2Y2	"	"	"
		"	85					GND					"				5.5 V				"		"	2Y3	"	"	"
		"	86			GND							"							5.5 V	"		"	2Y4	"	"	"
	I_{CCH}	3005	87	GND	5.5 V		5.5 V		5.5 V		5.5 V		"	5.5 V		5.5 V		5.5 V		5.5 V			"	V_{CC}		27	"
	I_{CCL}	"	88	GND	GND		GND		GND		GND		"	GND		GND		GND		GND			"	V_{CC}	-	46	"
	I_{CCZ}	"	89	5.5 V									"									7/	"	V_{CC}		54	"
	V_{IC}		90	-18 mA									"										4.5 V	1G		-1.5	V
			91		-18 mA								"										"	1A1		"	"

See footnotes at end of device types 02 and 03.

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TABLE III. Group A inspection for device types 02 and 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2,X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				1G	1A1	2Y4	1A2	2Y3	1A3	2Y2	1A4	2Y1	GND	2A1	1Y4	2A2	1Y3	2A3	1Y2	2A4	1Y1	2G 1/	V_{CC}				
1	V_{IC}		92				-18 mA						GND										4.5 V	1A2		-1.5	V
T_C = +25°C			93						-18 mA				"										"	1A3		"	"
			94								-18 mA		"										"	1A4		"	"
			95										"	-18 mA									"	2A1		"	"
			96										"			-18 mA							"	2A2		"	"
			97										"					-18 mA					"	2A3		"	"
			98										"							-18 mA			"	2A4		"	"
			99										"									-18 mA	"	2G		"	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																										
7 8/	Truth table tests	3014	100	B	B	L	B	L	B	L	B	L	GND	B	L	B	L	B	L	B	L	9/	5.0 V	10/			
T_C = +25°C		3014	101	B	A	H	A	H	A	H	A	H	GND	A	H	A	H	A	H	A	H	9/	5.0 V				
9	t_{PLH1}	3003 See fig. 4	102	GND	IN								GND								OUT		5.0 V	1A1 to 1Y1	2	18	ns
T_C = +25°C			103	"			IN						"						OUT				"	1A2 to 1Y2	"	"	"
			104	"					IN				"				OUT						"	1A3 to 1Y3	"	"	"
			105	"							IN		"		OUT								"	1A4 to 1Y4	"	"	"
			106									OUT	"	IN								6/	"	2A1 to 2Y1	"	"	"
			107							OUT			"			IN						"	"	2A2 to 2Y2	"	"	"
			108					OUT					"					IN				"	"	2A3 to 2Y3	"	"	"
			109			OUT							"							IN		"	"	2A4 to 2Y4	"	"	"
	t_{PHL1}	"	110	GND	IN								"								OUT		"	1A1 to 1Y1	"	"	"
		"	111	"			IN						"						OUT				"	1A2 to 1Y2	"	"	"
		"	112	"					IN				"				OUT						"	1A3 to 1Y3	"	"	"
		"	113	"							IN		"		OUT								"	1A4 to 1Y4	"	"	"
		"	114									OUT	"	IN								6/	"	2A1 to 2Y1	"	"	"
		"	115							OUT			"			IN						"	"	2A2 to 2Y2	"	"	"
		"	116					OUT					"					IN				"	"	2A3 to 2Y3	"	"	"
		"	117			OUT							"							IN		"	"	2A4 to 2Y4	"	"	"

See footnotes at end of device types 02 and 03.

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TABLE III. Group A inspection for device types 02 and 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2,X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				1G	1A1	2Y4	1A2	2Y3	1A3	2Y2	1A4	2Y1	GND	2A1	1Y4	2A2	1Y3	2A3	1Y2	2A4	1Y1	2G 1/	V_{CC}				
9	t_{PZL}	3003	118	IN	GND								GND								OUT		5.0 V	1G to 1Y1	2	30	ns
T_C = +25°C		See fig. 4	119	"			GND						"						OUT				"	1G to 1Y2	"	"	"
		"	120	"					GND				"				OUT						"	1G to 1Y3	"	"	"
		"	121	"							GND		"		OUT								"	1G to 1Y4	"	"	"
		"	122									OUT	"	GND								IN	"	2G to 2Y1	"	"	"
		"	123							OUT			"			GND						"	"	2G to 2Y2	"	"	"
		"	124					OUT					"					GND				"	"	2G to 2Y3	"	"	"
		"	125			OUT							"							GND		"	"	2G to 2Y4	"	"	"
	t_{PZH}	"	126	IN	4.5 V								"								OUT		"	1G to 1Y1	"	"	"
		"	127	"			4.5 V						"						OUT				"	1G to 1Y2	"	"	"
		"	128	"					4.5 V				"				OUT						"	1G to 1Y3	"	"	"
		"	129	"							4.5 V		"		OUT								"	1G to 1Y4	"	"	"
		"	130									OUT	"	4.5 V								IN	"	2G to 2Y1	"	"	"
		"	131							OUT			"			4.5 V						"	"	2G to 2Y2	"	"	"
		"	132					OUT					"					4.5 V				"	"	2G to 2Y3	"	"	"
		"	133			OUT							"							4.5 V		"	"	2G to 2Y4	"	"	"
	t_{PLZ}	"	134	IN	GND								"								OUT		"	1G to 1Y1	"	"	"
		"	135	"			GND						"						OUT				"	1G to 1Y2	"	"	"
		"	136	"					GND				"				OUT						"	1G to 1Y3	"	"	"
		"	137	"							GND		"		OUT								"	1G to 1Y4	"	"	"
		"	138									OUT	"	GND								IN	"	2G to 2Y1	"	"	"
		"	139							OUT			"			GND						"	"	2G to 2Y2	"	"	"
		"	140					OUT					"					GND				"	"	2G to 2Y3	"	"	"
		"	141			OUT							"							GND		"	"	2G to 2Y4	"	"	"
	t_{PHZ}	"	142	IN	4.5 V								"								OUT		"	1G to 1Y1	"	35	"
		"	143	"			4.5 V						"						OUT				"	1G to 1Y2	"	"	"
		"	144	"					4.5 V				"				OUT						"	1G to 1Y3	"	"	"
		"	145	"							4.5 V		"		OUT								"	1G to 1Y4	"	"	"
		"	146									OUT	"	4.5 V								IN	"	2G to 2Y1	"	"	"
		"	147							OUT			"			4.5 V						"	"	2G to 2Y2	"	"	"
		"	148					OUT					"					4.5 V				"	"	2G to 2Y3	"	"	"
		"	149			OUT							"							4.5 V		"	"	2G to 2Y4	"	"	"

See footnotes at end of device types 02 and 03.

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TABLE III. Group A inspection for device types 02 and 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2,X Test no.	1 IG	2 1A1	3 2Y4	4 1A2	5 2Y3	6 1A3	7 2Y2	8 1A4	9 2Y1	10 GND	11 2A1	12 1Y4	13 2A2	14 1Y3	15 2A3	16 1Y2	17 2A4	18 1Y1	19 2G 1/	20 V_{CC}	Measured terminal	Test Limits Min	Max	Unit
10 T_C = +125°C	t_{PLH1}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																						2	23	ns
	t_{PHL1}																							"	23	"
	t_{PZL}																							"	39	"
	t_{PZH}																							"	39	"
	t_{PLZ}																							"	39	"
	t_{PHZ}																							"	45	"
11 T_C = -55°C	-	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C																								

1/ For type 02, pin 19 is 2G; for type 03, pin 19 is 2C.
2/ Apply 2.0 V for type 02; apply 0.7 V for type 03.
3/ Apply 2.0 V for type 02; apply 0.5 V for type 03.
4/ Apply 0.7 V for type 02; apply 2.0 V for type 03.
5/ I_{IL} limits shall be as follows:

TEST	Min/Max limits (μA) for circuit A	B	C	D	E	F	G
I_{IL}	-5/-200	0/-150	0/-150	0/-150	-0/-200	-5/-200	0/-150

6/ Apply 5.5 V for type 02; apply GND for type 03.
7/ Apply GND for type 02; apply 5.5 V for type 03.
8/ A = 3.0 V minimum; B = 0.0 V or GND.
9/ Apply A for type 02; apply B for type 03.
10/ H >1.5 V; L <1.5 V.

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TABLE III. Group A inspection for device types 04 and 05.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2,X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limit Min	Test limit Max	Unit
				$1\overline{G}$	A1	A2	A3	A4	A5	A6	A7	A8	GND	Y8	Y7	Y6	Y5	Y4	Y3	Y2	Y1	$2\overline{G}$	V_{CC}				
1	V_{OH1}	3006	1	0.7 V	1/								GND								-3 mA	0.7 V	4.5 V	Y1	2.4		V
T_C = +25°C		"	2	"		1/							"							-3 mA		"	"	Y2	"		"
		"	3	"			1/						"						-3 mA			"	"	Y3	"		"
		"	4	"				1/					"					-3 mA				"	"	Y4	"		"
		"	5	"					1/				"				-3 mA					"	"	Y5	"		"
		"	6	"						1/			"			-3 mA						"	"	Y6	"		"
		"	7	"							1/		"		-3 mA							"	"	Y7	"		"
		"	8	"								1/	"	-3 mA								"	"	Y8	"		"
	V_{OH2}	"	9	0.5 V	2/								"								-12 mA	0.5 V	"	Y1	2.0		"
		"	10	"		2/							"							-12 mA		"	"	Y2	"		"
		"	11	"			2/						"						-12 mA			"	"	Y3	"		"
		"	12	"				2/					"					-12 mA				"	"	Y4	"		"
		"	13	"					2/				"				-12 mA					"	"	Y5	"		"
		"	14	"						2/			"			-12 mA						"	"	Y6	"		"
		"	15	"							2/		"		-12 mA							"	"	Y7	"		"
		"	16	"								2/	"	-12 mA								"	"	Y8	"		"
	V_{OL1}	3007	17	0.7 V	3/								"								12 mA	0.7 V	"	Y1		0.4	"
		"	18	"		3/							"							12 mA		"	"	Y2		"	"
		"	19	"			3/						"						12 mA			"	"	Y3		"	"
		"	20	"				3/					"					12 mA				"	"	Y4		"	"
		"	21	"					3/				"				12 mA					"	"	Y5		"	"
		"	22	"						3/			"			12 mA						"	"	Y6		"	"
		"	23	"							3/		"		12 mA							"	"	Y7		"	"
		"	24	"								3/	"	12 mA								"	"	Y8		"	"
	V_{OL2}	"	25	"	3/								"								18 mA	"	"	Y1		0.45	"
		"	26	"		3/							"							18 mA		"	"	Y2		"	"
		"	27	"			3/						"						18 mA			"	"	Y3		"	"
		"	28	"				3/					"					18 mA				"	"	Y4		"	"
		"	29	"					3/				"				18 mA					"	"	Y5		"	"
		"	30	"						3/			"			18 mA						"	"	Y6		"	"
		"	31	"							3/		"		18 mA							"	"	Y7		"	"
		"	32	"								3/	"	18 mA								"	"	Y8		"	"

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,Z,X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	-	Test limits		Unit
				$\overline{OE}$	A1	A2	A3	A4	A5	A6	A7	A8	GND	Y8	Y7	Y6	Y5	Y4	Y3	Y2	Y1	$\overline{2G}$	V_{CC}	Measured terminal	Min	Max	
1	I_{OZH}		33	2.0 V									GND								2.7 V		5.5 V	Y1		20	µA
T_C = +25°C			34	"									"							2.7 V			"	Y2		"	"
			35	"									"						2.7 V				"	Y3		"	"
			36	"									"					2.7 V					"	Y4		"	"
			37	"									"				2.7 V						"	Y5		"	"
			38	"									"			2.7 V							"	Y6		"	"
			39	"									"		2.7 V								"	Y7		"	"
			40	"									"	2.7 V									"	Y8		"	"
			41										"								2.7 V	2.0 V		"	Y1		"	"
			42										"							2.7 V			"	Y2		"	"
			43										"						2.7 V				"	Y3		"	"
			44										"					2.7 V					"	Y4		"	"
			45										"				2.7 V						"	Y5		"	"
			46										"			2.7 V							"	Y6		"	"
			47										"		2.7 V								"	Y7		"	"
			48										"	2.7 V									"	Y8		"	"
	I_{OZL}		49	2.0 V									"								0.4 V		"	Y1	-20	"	"
			50	"									"							0.4 V			"	Y2	"	"	"
			51	"									"						0.4 V				"	Y3	"	"	"
			52	"									"					0.4 V					"	Y4	"	"	"
			53	"									"				0.4 V						"	Y5	"	"	"
			54	"									"			0.4 V							"	Y6	"	"	"
			55	"									"		0.4 V								"	Y7	"	"	"
			56	"									"	0.4 V									"	Y8	"	"	"
			57										"								0.4 V	2.0 V		"	Y1	"	"	"
			58										"							0.4 V			"	Y2	"	"	"
			59										"						0.4 V				"	Y3	"	"	"
			60										"					0.4 V					"	Y4	"	"	"
			61										"				0.4 V						"	Y5	"	"	"
			62										"			0.4 V							"	Y6	"	"	"
			63										"		0.4 V								"	Y7	"	"	"
			64										"	0.4 V									"	Y8	"	"	"

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2,X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				1$\overline{G}$	A1	A2	A3	A4	A5	A6	A7	A8	GND	Y8	Y7	Y6	Y5	Y4	Y3	Y2	Y1	2$\overline{G}$	V_{CC}				
1	I_{IL}	3009	65	0.4 V									GND										5.5 V	1$\overline{G}$	4/	4/	µA
T_C = +25°C		"	66		0.4 V								"										"	A1	"	"	"
		"	67			0.4 V							"										"	A2	"	"	"
		"	68				0.4 V						"										"	A3	"	"	"
		"	69					0.4 V					"										"	A4	"	"	"
		"	70						0.4 V				"										"	A5	"	"	"
		"	71							0.4 V			"										"	A6	"	"	"
		"	72								0.4 V		"										"	A7	"	"	"
		"	73									0.4 V	"										"	A8	"	"	"
		"	74										"									0.4 V	"	2$\overline{G}$	"	"	"
	I_{IH1}	3010	75	2.7 V									"										"	1$\overline{G}$		20	"
		"	76		2.7 V								"										"	A1		"	"
		"	77			2.7 V							"										"	A2		"	"
		"	78				2.7 V						"										"	A3		"	"
		"	79					2.7 V					"										"	A4		"	"
		"	80						2.7 V				"										"	A5		"	"
		"	81							2.7 V			"										"	A6		"	"
		"	82								2.7 V		"										"	A7		"	"
		"	83									2.7 V	"										"	A8		"	"
		"	84										"									2.7 V	"	2$\overline{G}$		"	"
	I_{IH2}	"	85	5.5 V									"										"	1$\overline{G}$		100	"
		"	86		5.5 V								"										"	A1		"	"
		"	87			5.5 V							"										"	A2		"	"
		"	88				5.5 V						"										"	A3		"	"
		"	89					5.5 V					"										"	A4		"	"
		"	90						5.5 V				"										"	A5		"	"
		"	91							5.5 V			"										"	A6		"	"
		"	92								5.5 V		"										"	A7		"	"
		"	93									5.5 V	"										"	A8		"	"
		"	94										"									5.5 V	"	2$\overline{G}$		"	"

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2,X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{1G}$	A1	A2	A3	A4	A5	A6	A7	A8	GND	Y8	Y7	Y6	Y5	Y4	Y3	Y2	Y1	$\overline{2G}$	V_{CC}				
1 T_C = +25°C	I_{OS}	3011	95	GND	5/								GND								GND	GND	5.5 V	Y1	-40	-225	mA
		"	96	"		5/							"							GND		"	"	Y2	"	"	"
		"	97	"			5/						"						GND			"	"	Y3	"	"	"
		"	98	"				5/					"					GND				"	"	Y4	"	"	"
		"	99	"					5/				"				GND					"	"	Y5	"	"	"
		"	100	"						5/			"			GND						"	"	Y6	"	"	"
		"	101	"							5/		"		GND							"	"	Y7	"	"	"
		"	102	"								5/	"	GND								"	"	Y8	"	"	"
	I_{CCH}	3009	103	"	5/	5/	5/	5/	5/	5/	5/	5/	"									"	"	V_{CC}		7/	"
	I_{CCL}	"	104	"	6/	6/	6/	6/	6/	6/	6/	6/	"									"	"	V_{CC}		8/	"
	I_{CCZ}	"	105	5.5 V									"									5.5 V	"	V_{CC}		9/	"
	V_{IC}		106	-18 mA									"										4.5 V	$\overline{1G}$		-1.5	V
			107		-18 mA								"										"	A1		"	"
			108			-18 mA							"										"	A2		"	"
			109				-18 mA						"										"	A3		"	"
			110					-18 mA					"										"	A4		"	"
			111						-18 mA				"										"	A5		"	"
			112							-18 mA			"										"	A6		"	"
			113								-18 mA		"										"	A7		"	"
			114									-18 mA	"										"	A8		"	"
			115										"									-18 mA	"	$\overline{2G}$		"	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except TC = +125°C and VIC tests are omitted.																										
3	Same tests, terminal conditions, and limits as for subgroup 1, except TC = -55°C and VIC tests are omitted.																										
7	Truth table tests	3014	116	B	12/	12/	12/	12/	12/	12/	12/	12/	GND	H	H	H	H	H	H	H	H	B	5.0 V	11/			
T_C = +25°C		"	117	B	13/	13/	13/	13/	13/	13/	13/	13/	"	L	L	L	L	L	L	L	L	B	"	"			

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2,X Test no.	1 $1\overline{G}$	2 A1	3 A2	4 A3	5 A4	6 A5	7 A6	8 A7	9 A8	10 GND	11 Y8	12 Y7	13 Y6	14 Y5	15 Y4	16 Y3	17 Y2	18 Y1	19 $2\overline{G}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9	t_{PLH1}	3003	118	GND	IN								"								OUT	GND	5.0 V	A1 to Y1	2	14	ns
T_C = +25°C		Fig. 4	119	"		IN							"							OUT		"	"	A2 to Y2	"	"	"
		"	120	"			IN						"						OUT			"	"	A3 to Y3	"	"	"
		"	121	"				IN					"					OUT				"	"	A4 to Y4	"	"	"
		"	122	"					IN				"				OUT					"	"	A5 to Y5	"	"	"
		"	123	"						IN			"			OUT						"	"	A6 to Y6	"	"	"
		"	124	"							IN		"		OUT							"	"	A7 to Y7	"	"	"
		"	125	"								IN	"	OUT								"	"	A8 to Y8	"	"	"
	t_{PHL1}	"	126	"	IN								"								OUT	"	"	A1 to Y1	"	18	"
		"	127	"		IN							"						OUT			"	"	A2 to Y2	"	"	"
		"	128	"			IN						"					OUT				"	"	A3 to Y3	"	"	"
		"	129	"				IN					"				OUT					"	"	A4 to Y4	"	"	"
		"	130	"					IN				"			OUT						"	"	A5 to Y5	"	"	"
		"	131	"						IN			"				f					"	"	A6 to Y6	"	"	"
		"	132	"							IN		"		OUT							"	"	A7 to Y7	"	"	"
		"	133	"								IN	"	OUT								"	"	A8 to Y8	"	"	"
	t_{PZL}	"	134	IN	14/								"								OUT	"	"	$1\overline{G}$ to Y1	"	30	"
		"	135	"		14/							"						OUT			"	"	$1\overline{G}$ to Y2	"	"	"
		"	136	"			14/						"					OUT				"	"	$1\overline{G}$ to Y3	"	"	"
		"	137	"				14/					"				OUT					"	"	$1\overline{G}$ to Y4	"	"	"
		"	138	"					14/				"			OUT						"	"	$1\overline{G}$ to Y5	"	"	"
		"	139	"						14/			"			OUT						"	"	$1\overline{G}$ to Y6	"	"	"
		"	140	"							14/		"		OUT							"	"	$1\overline{G}$ to Y7	"	"	"
		"	141	"								14/	"	OUT								"	"	$1\overline{G}$ to Y8	"	"	"
		"	142	GND	14/								"								OUT	IN	"	$2\overline{G}$ to Y1	"	"	"
		"	143	"		14/							"						OUT			"	"	$2\overline{G}$ to Y2	"	"	"
		"	144	"			14/						"					OUT				"	"	$2\overline{G}$ to Y3	"	"	"
		"	145	"				14/					"				OUT					"	"	$2\overline{G}$ to Y4	"	"	"
		"	146	"					14/				"			OUT						"	"	$2\overline{G}$ to Y5	"	"	"
		"	147	"						14/			"			OUT						"	"	$2\overline{G}$ to Y6	"	"	"
		"	148	"							14/		"		OUT							"	"	$2\overline{G}$ to Y7	"	"	"
		"	149	"								14/	"	OUT								"	"	$2\overline{G}$ to Y8	"	"	"

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2,X1 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				1$\overline{G}$	A1	A2	A3	A4	A5	A6	A7	A8	GND	Y8	Y7	Y6	Y5	Y4	Y3	Y2	Y1	2$\overline{G}$	V_{CC}				
9	t_{PZH}	3003	150	IN	15/								GND								OUT	GND	5.0 V	1$\overline{G}$ to Y1	2	30	ns
T_C = +25°C		Fig. 4	151	"		15/							"							OUT		"	"	1$\overline{G}$ to Y2	"	"	"
		"	152	"			15/						"						OUT			"	"	1$\overline{G}$ to Y3	"	"	"
		"	153	"				15/					"					OUT				"	"	1$\overline{G}$ to Y4	"	"	"
		"	154	"					15/				"				OUT					"	"	1$\overline{G}$ to Y5	"	"	"
		"	155	"						15/			"			OUT						"	"	1$\overline{G}$ to Y6	"	"	"
		"	156	"							15/		"		OUT							"	"	1$\overline{G}$ to Y7	"	"	"
		"	157	"								15/	"	OUT								"	"	1$\overline{G}$ to Y8	"	"	"
		"	158	GND	15/								"								OUT	IN	"	2$\overline{G}$ to Y1	"	"	"
		"	159	"		15/							"							OUT		"	"	2$\overline{G}$ to Y2	"	"	"
		"	160	"			15/						"						OUT			"	"	2$\overline{G}$ to Y3	"	"	"
		"	161	"				15/					"					OUT				"	"	2$\overline{G}$ to Y4	"	"	"
		"	162	"					15/				"				OUT					"	"	2$\overline{G}$ to Y5	"	"	"
		"	163	"						15/			"			OUT						"	"	2$\overline{G}$ to Y6	"	"	"
		"	164	"							15/		"		OUT							"	"	2$\overline{G}$ to Y7	"	"	"
		"	165	"								15/	"	OUT								"	"	2$\overline{G}$ to Y8	"	"	"
	t_{PLZ}	3003	166	IN	14/								"								OUT	GND	"	1$\overline{G}$ to Y1	"	"	"
		Fig. 4	167	"		14/							"							OUT		"	"	1$\overline{G}$ to Y2	"	"	"
		"	168	"			14/						"						OUT			"	"	1$\overline{G}$ to Y3	"	"	"
		"	169	"				14/					"					OUT				"	"	1$\overline{G}$ to Y4	"	"	"
		"	170	"					14/				"				OUT					"	"	1$\overline{G}$ to Y5	"	"	"
		"	171	"						14/			"			OUT						"	"	1$\overline{G}$ to Y6	"	"	"
		"	172	"							14/		"		OUT							"	"	1$\overline{G}$ to Y7	"	"	"
		"	173	"								14/	"	OUT								"	"	1$\overline{G}$ to Y8	"	"	"
		"	174	GND	14/								"								OUT	IN	"	2$\overline{G}$ to Y1	"	"	"
		"	175	"		14/							"							OUT		"	"	2$\overline{G}$ to Y2	"	"	"
		"	176	"			14/						"						OUT			"	"	2$\overline{G}$ to Y3	"	"	"
		"	177	"				14/					"					OUT				"	"	2$\overline{G}$ to Y4	"	"	"
		"	178	"					14/				"				OUT					"	"	2$\overline{G}$ to Y5	"	"	"
		"	179	"						14/			"									"	"	2$\overline{G}$ to Y6	"	"	"
		"	180	"							14/		"		OUT									2$\overline{G}$ to Y7	"	"	"
		"	181	"								14/	"	OUT								"	"	2$\overline{G}$ to Y8	"	"	"

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2,X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				1G	A1	A2	A3	A4	A5	A6	A7	A8	GND	Y8	Y7	Y6	Y5	Y4	Y3	Y2	Y1	2G	V_{CC}				
9	t_{PHZ}	3003	182	IN	15/								GND							OUT	OUT	GND	5.0 V	1G to Y1	2	35	ns
T_C = +25°C		Fig. 4	183	"		15/							"						OUT			"	"	1G to Y2	"	"	"
		"	184	"			15/						"					OUT				"	"	1G to Y3	"	"	"
		"	185	"				15/					"				OUT					"	"	1G to Y4	"	"	"
		"	186	"					15/				"			OUT						"	"	1G to Y5	"	"	"
		"	187	"						15/			"		OUT							"	"	1G to Y6	"	"	"
		"	188	"							15/		"	OUT								"	"	1G to Y7	"	"	"
		"	189	"								15/	"								OUT	IN	"	1G to Y8	"	"	"
		"	190	GND	15/								"							OUT		"	"	2G to Y1	"	"	"
		"	191	"		15/							"						OUT			"	"	2G to Y2	"	"	"
		"	192	"			15/						"					OUT				"	"	2G to Y3	"	"	"
		"	193	"				15/					"				OUT					"	"	2G to Y4	"	"	"
		"	194	"					15/				"			OUT						"	"	2G to Y5	"	"	"
		"	195	"						15/			"		OUT							"	"	2G to Y6	"	"	"
		"	196	"							15/		"									"	"	2G to Y7	"	"	"
		"	197	"								15/	"	OUT								"	"	2G to Y8	"	"	"
10	t_{PLH1}																								"	18	"
T_C = +125°C	t_{PHL1}																								"	23	"
	t_{PZL}																								"	39	"
	t_{PZH}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																							"	"	"
	t_{PLZ}																								"	"	"
	t_{PHZ}																								"	45	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																										

1/ For device type 04; 0.7 V
For device type 05; 2.0 V.
2/ For device type 04; 0.5 V
For device type 05; 2.0 V.
3/ For device type 04; 2.0 V
For device type 05; 0.7 V.
4/ I_{IL} limits shall be as follows:

Test	Min/max limits (µA) for circuit A	B	C	D	E	F	G
I_{IL}	0/-200						

5/ For device type 04; GND
For device type 05; 5.5 V.
6/ For device type 04; 5.5 V
For device type 05; GND.
7/ For device type 04; 27 mA
For device type 05; 32 mA.
8/ For device type 04; 45 mA
For device type 05; 52 mA.
9/ For device type 04; 52 mA
For device type 05; 55 mA.
10/ A = 3.0 V min; B = 0.0 V or GND.
11/ H >1.5 V; L <1.5 V.
12/ For device type 04; use B
For device type 05; use A.
13/ For device type 04; use A
For device type 05; use B.
14/ For device type 04; 4.5 V
For device type 05; GND.
15/ For device type 04; GND
For device type 05; 4.5 V.

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, 6, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection parameters for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

c. Steady-state life-test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D, E, or F (condition F applies to class B only) using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.6 Inclusion with other detail specifications. For qualification and quality conformance inspection purposes, devices covered by this specification may be treated as though they were included on the same detail specification as devices covered by MIL-M-38510/3288. In addition, if a manufacturer is already part I qualified for type 32802, and if the respective devices on this specification (MIL-M-38510/324C) are designed and manufactured identically (same die, same process, same screening) in all respects except electrical testing, then device type 32402 may be part I qualified by conducting only group A electrical tests with approval of the qualifying activity including subgroups A-10 and A-11, and submitting data in accordance with MIL-M-38510, appendix D (i.e., groups B, C, and D tests are not required).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be as specified in MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

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6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable, These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
V_{IN}	Voltage level at an input terminal
I_{IN}	Current flowing into an input terminal
t_{PHZ}	Output disable time from high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined high level to a high-impedance (off) state.
t_{PLZ}	Output disable time from low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined low level to a high-impedance (off) state.
t_{PZH}	Output enable time to high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high impedance (off) state to the defined high level.
t_{PZL}	Output enable time to low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined low level.
I_{OZL}	Output current in the high-impedance mode with the output voltage low.
I_{OZH}	Output current in the high-impedance mode with the output voltage high.

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6.4 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer lead lengths and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS240
02	54LS241
03	54LS244
04	54LS540
05	54LS541

6.6 Manufacturers' designations. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuit						
	A	B	C	D	E	F	G
	Texas Instruments	Signetics Corp.	National Semiconductor Corp.	Raytheon Co.	Fairchild Semiconductor	Motorola Inc.	Advanced Micro Devices
01	X	X	X	X	X	X	X
02	X	X	X	X	X	X	X
03	X	X	X	X	X	X	X
04	X						
05	X						

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1021)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/324C	MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY, TTL, OCTAL BUFFER GATES WITH THREE-STATE OUTPUTS,

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
MONOLITHIC SILICON	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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